

Common Household Chemicals and the Allergy Risks in Pre-School Age Children

Background The risk of indoor exposure to volatile organic compounds (VOCs) on allergic airway diseases in children remains unknown. **Objective** We examined the residential concentrations of VOCs, emitted from building materials, paints, furniture, and other lifestyle practices and the risks of multiple allergic diseases as well as the IgE-sensitization in pre-school age children in Sweden. **Methods** In a case-control investigation (198 case children with asthma and allergy and 202 healthy controls), air samples were collected in the room where the child slept. The air samples were analyzed for the levels of eight classes of VOCs. **Results** A natural-log unit of summed propylene glycol and glycol ethers (PGEs) in bedroom air (equal to interquartile range, or 3.43 – 15.65 $\mu\text{g}/\text{m}^3$) was associated with 1.5-fold greater likelihood of being a case (95% CI, 1.1 – 2.1), 1.5-fold greater likelihood of asthma (95% CI, 1.0 – 2.3), 2.8-fold greater likelihood of rhinitis (95% CI, 1.6 – 4.7), and 1.6-fold greater likelihood of eczema (95% CI, 1.1 – 2.3), accounting for gender, secondhand smoke, allergies in both parents, wet cleaning with chemical agents, construction period of the building, limonene, cat and dog allergens, butyl benzyl phthalate (BBzP), and di(2-ethylhexyl)phthalate (DEHP). When the analysis was restricted to the cases, the same unit concentration was associated with 1.8-fold greater likelihood of IgE-sensitization (95% CI, 1.1 – 2.8) compared to the non-IgE sensitized cases. No similar associations were found for the other classes of VOCs. **Conclusion** We propose a novel hypothesis that PGEs in indoor air exacerbate and/or induce the multiple allergic symptoms, asthma, rhinitis and eczema, as well as IgE sensitization respectively.